

Unifying Special Relativity and Quantum Mechanics

Lecture 5 — Unitarity of the Massive Irreps, the Spin–Polarization Basis, and the Massless Case: Helicity

(Typed version of the handwritten notes, with added explanations in blue boxes)

Added explanation (not in the original notes)

How to read this document. Same conventions as Lectures 1–4: everything in the normal text and in the yellow boxes is a faithful transcription of the handwritten notes (yellow boxes correspond to the results that were highlighted in the original — in this lecture some of them are highlighted in pink/red rather than yellow, but I use a single yellow style for all of them); everything inside a blue box like this one is extra material that I added. The figures are redrawn versions of the hand sketches; figures that have no handwritten counterpart are explicitly marked as added.

Two remarks on the pagination of this scan (same situation as in Lecture 3): (i) the first four scanned pages carry the numbers 29–32, i.e. they are pages [p. 29*]–[p. 32*] of the second (starred) numbering sequence and *repeat the final pages of Lecture 4* (factorization of $U(\Lambda, a)$, action of translations, action of Lorentz transformations, and the Wigner-rotation formula). Since they were already typed at the end of Lecture 4, this document starts from the first genuinely new page, [p. 33*]. (ii) After [p. 45*] (and one blank page) the notes open a new chapter, “Unitary irreps of the Poincaré group — massless case”, and restart the page numbering from 1 *again*. To avoid confusion with the pages of Lecture 1 and with the starred pages of Lectures 3–5, the markers of that part are written [p. 1**], ..., [p. 6**].

Where we left off (recap of Lecture 4). For the massive case ($p^2 = M^2 > 0$, $p^0 > 0$) we built the states $|M, \vec{p}, s, \lambda\rangle \equiv U_H(\hat{p}, |\vec{p}|/M)|M, \vec{0}, s, \lambda\rangle$ using the standard boost $H(\hat{p}, |\vec{p}|/M) = \Lambda_R(\hat{z} \rightarrow \hat{p}) \Lambda_B(\eta, \hat{z})$, we showed that λ is the *helicity* ($\vec{J} \cdot \vec{P} |M, \vec{p}, s, \lambda\rangle = \lambda |\vec{p}| |M, \vec{p}, s, \lambda\rangle$), and we derived the transformation law

$$U(\Lambda)|M, \vec{p}, s, \lambda\rangle = \sum_{\lambda'} [D_s(W(\Lambda, p))]_{\lambda'\lambda} |M, \vec{p}_\Lambda, s, \lambda'\rangle,$$

$$W(\Lambda, p) = H(\hat{p}_\Lambda, \frac{|\vec{p}_\Lambda|}{M})^{-1} \Lambda H(\hat{p}, \frac{|\vec{p}|}{M}) \in SO(3),$$

with $\vec{p}_\Lambda$ the spatial part of $p_\Lambda \equiv \Lambda p$. Three checks were announced (p. 28*): (1) minimal invariance of the subspace $\{|M, \vec{p}, s, \lambda\rangle\}_{M,s}$; (2) the Poincaré composition rule; (3) unitarity. Check (1) is exactly the content of the boxed formula above; this lecture completes checks (2) and (3), then re-does Step 5 with a different (rotation-friendly) choice of standard boost, and finally opens the *massless* case, where the little group turns out to be our old friend $SE(2)$ and the helicity emerges as the only spin-like label.

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1 Composition rule for the massive irrep [p. 33*]

Given this action, it is possible to prove that

$$U(\Lambda_2)U(\Lambda_1) = U(\Lambda_2\Lambda_1)$$

$\hookrightarrow$ Intuition behind: we reduced the action of a generic Lorentz transformation to the action of ordinary rotations, and the matrices $D_s(W(\Lambda, p))$ do respect the composition rule of rotations (since D_s form a representation).

More in general, we also have

$$U(\Lambda_2, a_2)U(\Lambda_1, a_1) = U(\Lambda_2\Lambda_1, \Lambda_2a_1 + a_2)$$

For a generic Poincaré, we get

$$U(\Lambda, a)|M, \vec{p}, s, \lambda\rangle = e^{ia \cdot p_\Lambda} \sum_{\lambda'} [D_s(W(\Lambda, p))]_{\lambda'\lambda} |M, \vec{p}_\Lambda, s, \lambda'\rangle$$

► first, we act with $U(\Lambda)$; then, we get the phase $\exp(ia \cdot p_\Lambda)$ acting with $U(a)$ on $|M, \vec{p}_\Lambda, s, \lambda'\rangle$.

Added explanation (not in the original notes)

Why the one-line “intuition” is actually the whole proof. Apply $U(\Lambda_2)U(\Lambda_1)$ to $|M, \vec{p}, s, \lambda\rangle$ using the boxed transformation law twice: the momentum goes $\vec{p} \rightarrow \vec{p}_{\Lambda_1} \rightarrow \vec{p}_{\Lambda_2\Lambda_1}$, and the spin index picks up $D_s(W(\Lambda_2, \Lambda_1 p)) D_s(W(\Lambda_1, p))$. On the other hand, $U(\Lambda_2\Lambda_1)$ produces

$D_s(W(\Lambda_2\Lambda_1, p))$. The two agree because of the *cocycle property* of the Wigner rotation,

$$W(\Lambda_2\Lambda_1, p) = W(\Lambda_2, \Lambda_1 p) W(\Lambda_1, p),$$

which follows in one line from the definition $W(\Lambda, p) = H(p_\Lambda)^{-1} \Lambda H(p)$ (shorthand $H(p) \equiv H(\vec{p}, |\vec{p}|/M)$):

$$W(\Lambda_2, \Lambda_1 p) W(\Lambda_1, p) = H(p_{\Lambda_2\Lambda_1})^{-1} \Lambda_2 \underbrace{H(p_{\Lambda_1}) H(p_{\Lambda_1})^{-1}}_{= \mathbb{1}} \Lambda_1 H(p) = H(p_{\Lambda_2\Lambda_1})^{-1} (\Lambda_2\Lambda_1) H(p),$$

and because D_s is a representation of the little group, $D_s(W_2)D_s(W_1) = D_s(W_2W_1)$. This was anticipated in the final blue box of Lecture 4; the notes state the result and give the intuition without writing the chain out.

For the full Poincaré rule, combine this with $U(a)U(\Lambda)$: acting on $|M, \vec{p}, s, \lambda\rangle$, first $U(\Lambda)$ moves the state to momentum p_Λ (rotating the spin index), and then the translation $U(a) = e^{ia_\mu P^\mu}$ sees an eigenstate of P^μ with eigenvalue p_Λ^μ , hence contributes the phase $e^{ia \cdot p_\Lambda}$ — exactly the boxed formula. Note the phase involves p_Λ , not p : the order of the factorization $U(\Lambda, a) = U(a)U(\Lambda)$ matters, as stressed since Lecture 1 (p. 8).

2 Unitarity [p. 33*–35*]

Unitarity. As a final step, let us consider unitarity. We introduce, for each subspace with s, M fixed, [p. 34*]

$$\langle M, \vec{p}, s, \lambda | M, \vec{q}, s, \lambda' \rangle = (2\pi)^3 \delta_{\lambda\lambda'} (2E_p) \delta(\vec{p} - \vec{q}), \quad \mathbb{1} = \sum_\lambda \int \frac{d^3\vec{p}}{(2\pi)^3 2E_p} |M, \vec{p}, s, \lambda\rangle \langle M, \vec{p}, s, \lambda|$$

with

$$E_p = (|\vec{p}|^2 + M^2)^{1/2}.$$

Added explanation (not in the original notes)

Why this normalization, and not the $\delta(\vec{p} - \vec{q})$ of Lecture 2? For $SE(2)$ we used $\langle \vec{p} | \vec{q} \rangle = \delta(\vec{p} - \vec{q})$ and $\int d^2p |\vec{p}\rangle \langle \vec{p}| = \mathbb{1}$, and unitarity of rotations worked because the measure d^2p is rotation invariant ($\det R = 1$). Under a *boost*, however, d^3p alone is *not* invariant (a boost squeezes momentum space), so that convention would make the completeness relation frame dependent. The fix is to insert the energy factor: the combination $d^3p/(2E_p)$ is Lorentz invariant. Cleanest way to see it: write it as a manifestly invariant four-dimensional integral restricted to the mass shell,

$$\int \frac{d^3p}{2E_p} f(p) = \int d^4p \delta(p^2 - M^2) \theta(p^0) f(p),$$

which holds because $\delta((p^0)^2 - E_p^2) = \frac{1}{2E_p} [\delta(p^0 - E_p) + \delta(p^0 + E_p)]$ and $\theta(p^0)$ keeps only the upper mass-shell branch. Every ingredient on the right (d^4p , since $|\det \Lambda| = 1$; p^2 ; the sign of p^0 on the time-like shell, by the connectedness argument of Lecture 3, p. 2*–3*) is invariant under $SO^\uparrow(1, 3)$ — hence so is the measure. The matching normalization of the states then has to be $(2\pi)^3 2E_p \delta(\vec{p} - \vec{q})$, which is the relativistic (“covariant”) normalization used throughout quantum

field theory. The factors of $(2\pi)^3$ are pure convention (they cancel between the inner product and the completeness relation).

It is now possible to check unitarity, $U(\Lambda, a) U(\Lambda, a)^\dagger = \mathbb{1}$:

$$U(\Lambda, a) U(\Lambda, a)^\dagger = \sum_{\lambda} \int \frac{d^3 \vec{p}}{(2\pi)^3 2E_p} U(\Lambda, a) |M, \vec{p}, s, \lambda\rangle \langle M, \vec{p}, s, \lambda| U(\Lambda, a)^\dagger$$

where, from the boxed transformation law,

$$U(\Lambda, a) |M, \vec{p}, s, \lambda\rangle = e^{ip_\Lambda \cdot a} \sum_{\lambda'} [D_s(W(\Lambda, p))]_{\lambda' \lambda} |M, \vec{p}_\Lambda, s, \lambda'\rangle,$$

$$\langle M, \vec{p}, s, \lambda| U(\Lambda, a)^\dagger = \sum_{\lambda''} \langle M, \vec{p}_\Lambda, s, \lambda''| e^{-ip_\Lambda \cdot a} [D_s(W(\Lambda, p))]_{\lambda'' \lambda}^*.$$

Substituting (the translation phases cancel):

$$= \sum_{\lambda} \int \frac{d^3 \vec{p}}{(2\pi)^3 2E_p} \sum_{\lambda', \lambda''} [D_s(W(\Lambda, p))]_{\lambda' \lambda} [D_s(W(\Lambda, p))]_{\lambda'' \lambda}^* |M, \vec{p}_\Lambda, s, \lambda'\rangle \langle M, \vec{p}_\Lambda, s, \lambda''|$$

with

$$D_s^\dagger = D_s^{-1} = (D_s^T)^*, \quad [D_s(W)]_{\lambda'' \lambda}^* = [D_s(W)^\dagger]_{\lambda \lambda''}.$$

[p. 35*]

$$\begin{aligned} &= \int \frac{d^3 \vec{p}}{(2\pi)^3 2E_p} \sum_{\lambda', \lambda''} \left[\underbrace{\sum_{\lambda} (D_s(W))_{\lambda' \lambda} (D_s(W)^\dagger)_{\lambda \lambda''}}_{\text{rotations are unitary: } = \delta_{\lambda' \lambda''}} \right] |M, \vec{p}_\Lambda, s, \lambda'\rangle \langle M, \vec{p}_\Lambda, s, \lambda''| \\ &= \underbrace{\int \frac{d^3 \vec{p}}{(2\pi)^3 2E_p}}_{\substack{\text{this measure is} \\ \text{Lorentz invariant}}} \sum_{\lambda} |M, \vec{p}_\Lambda, s, \lambda\rangle \langle M, \vec{p}_\Lambda, s, \lambda| \quad \begin{array}{l} \text{we now make a change of variables,} \\ \vec{p}_\Lambda = \Lambda p = \vec{k} \rightsquigarrow p = \Lambda^{-1} k \end{array} \\ &= \int \frac{d^3 \vec{k}}{(2\pi)^3 2E_k} \sum_{\lambda} |M, \vec{k}, s, \lambda\rangle \langle M, \vec{k}, s, \lambda| = \mathbb{1}. \end{aligned}$$

Added explanation (not in the original notes)

Logic of the proof — same skeleton as the $SE(2)$ unitarity proof of Lecture 2 (p. 26), with two upgrades. (i) There, the group acted on the momentum label only, and invariance of the measure ($d^2 p' = d^2 p$) closed the argument. Here the group also acts on the discrete spin label, and that action is taken care of by the unitarity of the little-group matrices: $\sum_{\lambda} (D_s)_{\lambda' \lambda} (D_s^\dagger)_{\lambda \lambda''} = (D_s D_s^\dagger)_{\lambda' \lambda''} = \delta_{\lambda' \lambda''}$ — this is where it pays off that the Wigner rotation is an honest rotation, represented by the *unitary* spin- s matrices D_s . (ii) The change of variables now involves a boost, and this is where the relativistic measure $d^3 p / (2E_p)$ earns its keep (previous blue box). The phases $e^{\pm ip_\Lambda \cdot a}$ from the translation part cancel between ket and bra, just as in Lecture 2. Strictly, one shows $UU^\dagger = \mathbb{1}$ and, by the same computation read backwards, $U^\dagger U = \mathbb{1}$; together with linearity this is unitarity. With this, all three checks of p. 28* are complete: the massive unitary irreps of the Poincaré group are fully constructed.

3 Comment: particles as irreps; helicity [p. 35*–36*]

Comment.

■ We associate to each unitary irrep of Poincaré a QM free relativistic particle with mass M and spin s .

↔ Such particle has $(2s + 1)$ degrees of freedom associated to the helicity quantum number.

■ Helicity is a good quantum number since it commutes with the Hamiltonian and they can be diagonalized together.

[p. 36*]

However, for a massive particle helicity is not invariant under Lorentz transformation.

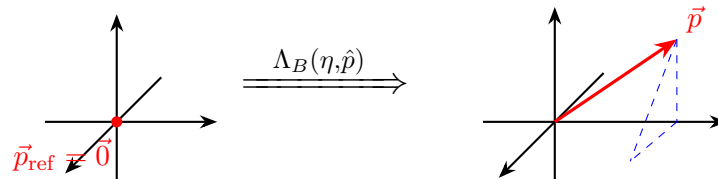
↔ note, however, that helicity *is* invariant under rotations.

Added explanation (not in the original notes)

Unpacking the three statements. (i) “*Commutates with the Hamiltonian*”: the helicity operator is (up to the normalization $1/|\vec{P}|$) $\vec{J} \cdot \vec{P}$, and $[\vec{J} \cdot \vec{P}, P^0] = \sum_k [J^k, P^0] P^k = 0$ by the Poincaré bracket $[J^k, P^0] = 0$ of Lecture 3 (p. 51); it also commutes with $\vec{P}$ and with $\vec{J} \cdot \hat{n}$ rotations about the momentum axis. So helicity, energy and momentum are simultaneously measurable — a good labelling of one-particle states. (ii) “*Invariant under rotations*”: under a rotation both $\vec{J}$ and $\vec{P}$ rotate as vectors, so the scalar product $\vec{J} \cdot \vec{P}$ is unchanged — a rotated observer assigns the same λ . (This will be re-derived group-theoretically in Sec. 7: $W(\Lambda_R, p) = \Lambda_R$ in the spin-polarization basis, and in the helicity basis the corresponding statement is that rotations do not mix helicities.) (iii) “*Not Lorentz invariant*” (*massive case*): a massive particle moves slower than light, so an observer can *overtake* it; in that observer’s frame $\vec{p}$ points the opposite way while the spin is unchanged, so $\lambda = \vec{J} \cdot \hat{p}$ flips sign. More generally a boost changes the direction of $\vec{p}$ but not of the spin quantization, mixing the $2s + 1$ helicity components. For a *massless* particle no observer can overtake it — and indeed at the end of this lecture helicity will emerge as a genuine Lorentz invariant labelling massless irreps.

4 Step 5, an alternative view: direct boost and the spin–polarization basis [p. 36*–38*]

Step 5: an alternative view. Go back at Step 5. In momentum space, we can also define



We can directly boost in direction $\hat{p}$:

$$p^\mu = \Lambda_B(\eta, \hat{p})^\mu{}_\nu p_{\text{ref}}^\nu = \begin{pmatrix} p^0 \\ |\vec{p}| \hat{p} \end{pmatrix}, \quad \cosh \eta = \frac{p^0}{M}, \quad \sinh \eta = \frac{|\vec{p}|}{M}.$$

► we can trade the rapidity for the ratio $|\vec{p}|/M$, and write

$$\Lambda_B(|\vec{p}|/M, \hat{p}) \equiv L(p) \quad (\text{Weinberg notation}).$$

[p. 37*] We remark that

$$\Lambda_B(\hat{p}, |\vec{p}|/M) \neq \Lambda_R(\hat{z} \rightarrow \hat{p}) \Lambda_B(|\vec{p}|/M, \hat{z})$$

► their effect on p_{ref} is the same, but they are different transformations.

What changes, if we make this choice in our construction?

1) We introduce the operator $U_B \equiv U_B(|\vec{p}|/M, \hat{p})$ which implements the action of the boost Λ_B on the reference state $|M, \vec{0}, s, \sigma\rangle$.

$$\Downarrow \quad \text{since} \quad P^\mu U_B |M, \vec{0}, s, \sigma\rangle = p^\mu U_B |M, \vec{0}, s, \sigma\rangle \quad \text{we define}$$

$$|M, \vec{p}, s, \sigma\rangle \equiv U_B(|\vec{p}|/M, \hat{p}) |M, \vec{0}, s, \sigma\rangle$$

2) The eigenvalue equations for P^μ , P^2 , W^2 remain the same (with eigenvalues p^μ , M^2 , $-M^2 s(s+1)$). What changes is the last equation.

Added explanation (not in the original notes)

Why the two transformations differ, and why it matters. Both $\Lambda_B(\hat{p}, |\vec{p}|/M)$ and $H(\hat{p}, |\vec{p}|/M) = \Lambda_R(\hat{z} \rightarrow \hat{p}) \Lambda_B(|\vec{p}|/M, \hat{z})$ map $p_{\text{ref}} = (M, \vec{0})$ to the same p^μ — but a Lorentz transformation is not determined by its action on a single vector: two maps with the same effect on p_{ref} differ by an element of the little group of p_{ref} (a rotation). Indeed $H^{-1} \Lambda_B(\hat{p}, \dots)$ fixes p_{ref} , so it is some rotation $R_0(\vec{p}) \in SO(3)$. Choosing the “standard boost” therefore really is a *convention* (as anticipated in the Lecture 4 blue box on p. 21*): the direct boost $L(p)$ (the choice made in Weinberg’s book, *The Quantum Theory of Fields*, Vol. 1) and the boost-then-rotate $H(p)$ of Lecture 4 generate two different bases of the *same* irrep, related state by state through the little-group matrix $D_s(R_0(\vec{p}))$. The physics (the pair (M, s) , the transformation law, all probabilities) is convention independent; what changes is the *meaning of the label* carried by the states — helicity in one basis, spin projection along a fixed axis in the other, as the notes now show. A practical rule of thumb: the helicity basis is the natural one for massless particles and for decay kinematics; the spin-polarization basis is the natural one for adding spins and for the non-relativistic limit (where it reduces to the usual $|s, \sigma\rangle$ of QM with $\hat{z}$ quantization axis).

[p. 38*] We define

$$S_B^\mu = \Lambda_B^\mu{}_\nu S^\nu, \quad S_{\hat{z}}^\mu = (\Lambda_B^{-1})^\mu{}_\nu S_B^\nu$$

and consider the operator $-S_B^\mu W_\mu / M$. We write

$$\begin{aligned} -\frac{S_B^\mu W_\mu}{M} |M, \vec{p}, s, \sigma\rangle &= -\frac{S_B^\mu W_\mu}{M} U_B(|\vec{p}|/M, \hat{p}) |M, \vec{0}, s, \sigma\rangle \\ &= -\frac{S_B^\mu}{M} U_B \underbrace{(U_B^{-1} W_\mu U_B)}_{(\Lambda_B)_\mu{}^\nu W_\nu = (\Lambda_B^{-1})^\nu{}_\mu W_\nu} |M, \vec{0}, s, \sigma\rangle \end{aligned}$$

$$\begin{aligned}
&= U_B \underbrace{\frac{(-S_B^\mu)(\Lambda_B^{-1})^\nu{}_\mu}{M}}_{-S_z^\nu/M} W_\nu |M, \vec{0}, s, \sigma\rangle = U_B \left(-\frac{S_z^\nu W_\nu}{M} \right) |M, \vec{0}, s, \sigma\rangle \\
&= U_B \sigma |M, \vec{0}, s, \sigma\rangle = \sigma |M, \vec{p}, s, \sigma\rangle.
\end{aligned}$$

$$-\frac{S_B^\mu W_\mu}{M} |M, \vec{p}, s, \sigma\rangle = \sigma |M, \vec{p}, s, \sigma\rangle$$

► what changes is the interpretation of the eigenvalue “ σ ”. “ σ ” is now eigenvalue of the operator $-S_B^\mu W_\mu/M$ (it is not the helicity anymore). What is, now, “ σ ”?

Added explanation (not in the original notes)

Reading the chain. It is the same computation done in Lecture 4 (p. 24*) for the helicity basis, with $U_H \rightarrow U_B$ and $S_p^\mu \rightarrow S_B^\mu$: sandwich W_μ between $U_B^{-1}(\dots)U_B$, use that W^μ is a four-vector of operators (Lecture 4, p. 17*–18*), so conjugation replaces it by the numerically boosted combination, and let the boost matrices eat the four-vector S_B back into $S_z = (0, 0, 0, 1)$, whose eigenvalue equation on the rest states is known ($-S_z \cdot W/M = “J^3”$ there, eigenvalue σ ; Lecture 4, p. 19*). The middle step uses the index identity $(\Lambda)_\mu{}^\nu = (\Lambda^{-1})^\nu{}_\mu$ from Lecture 2 (p. 32). Note that Λ_B here is a fixed numerical matrix (it depends on the c-number $\vec{p}$ labelling the state, not on the operator $\vec{P}$), so it can be moved freely through the operators.

5 The spin operator $\vec{S}$ [p. 39*–40*]

[p. 39*] Let us write

$$-\frac{S_B^\mu W_\mu}{M} = -\frac{\Lambda_B^\mu{}_\nu S_z^\nu W_\mu}{M} = -\frac{1}{M} (\Lambda_B)^\mu{}_3 W_\mu = +\frac{1}{M} (\Lambda_B)_\mu{}^3 W^\mu = \frac{1}{M} (\Lambda_B^{-1})^3{}_\mu W^\mu.$$

We then define the operators

$$S^k \equiv \frac{1}{M} (\Lambda_B^{-1})^k{}_\mu W^\mu \quad (\otimes)$$

such that the eigenvalue equation reads

$$S^3 |M, \vec{p}, s, \sigma\rangle = \sigma |M, \vec{p}, s, \sigma\rangle$$

The three operators S^k enjoy the following properties:

1)

$$[S^k, P^i] = 0$$

► this is a direct consequence of $[W^\mu, P^\nu] = 0$.

2)

$$[S^i, S^j] = i \varepsilon_{ijk} S^k$$

The most intuitive proof is the following. From the definition $(\otimes)$ we see that, since W^μ is a 4-vector, $(\Lambda_B^{-1})^\mu_k$ acting on it transforms it back to [p. 40*] the rest frame:

$$(\Lambda_B^{-1})^\mu_k W^\mu = W_{\text{rest}}^k.$$

► since $\vec{W} = \vec{J}P^0 + \vec{K} \times \vec{P}$, we have $\vec{W}_{\text{rest}} = M\vec{J}$.

$$[S^i, S^j] = \frac{1}{M^2} [MJ^i, MJ^j] = [J^i, J^j] = i \varepsilon_{ijk} J^k = i \varepsilon_{ijk} \frac{1}{M} (\Lambda_B^{-1})^\mu_k W^\mu = i \varepsilon_{ijk} S^k.$$

We thus conclude that $\vec{S}$ corresponds to the spin angular momentum.

We thus have

$$\left\{ |M, \vec{p}, s, \lambda\rangle \right\} \quad \text{“helicity basis”; } \lambda \text{ is the helicity}$$

► projection of the spin along the direction of motion: λ .

$$\left\{ |M, \vec{p}, s, \sigma\rangle \right\} \quad \text{“spin-polarization basis”}$$

► projection of the spin along the fixed $\hat{z}$ direction: σ .

Added explanation (not in the original notes)

What the operators S^k really are, and a caveat on the “intuitive proof”. The definition $(\otimes)$ says: take the Pauli–Lubanski four-vector on a state of momentum $\vec{p}$, and apply the *inverse* of the standard boost that created that state — i.e. read off the spin *in the particle’s own rest frame*, reached by the conventional boost. That is why $\vec{S}$ deserves the name *spin operator*: it is the rest-frame angular momentum, transported to the moving states by the same convention that defines them. On the reference subspace $(\vec{p} = \vec{0}, \Lambda_B = \mathbb{1})$ one has $S^k = W^k/M = J^k$ exactly.

The chain written in the notes for property 2) is a physicist’s shorthand: the matrices $(\Lambda_B^{-1})^\mu_k$ depend on $\vec{p}$, so the substitution $(\Lambda_B^{-1})W = W_{\text{rest}} = M\vec{J}$ and the bracket evaluation are to be understood *at fixed momentum*, i.e. acting on the momentum eigenstates $|M, \vec{p}, s, \sigma\rangle$, on which $\vec{p}$ is a c-number. This is legitimate because $[S^k, P^i] = 0$ (property 1): the S^k do not move states between momentum eigenspaces, so the algebra can be checked eigenspace by eigenspace, and on each eigenspace Λ_B is a constant matrix and W^μ acts exactly like the rest-frame $M(0, \vec{J})$ conjugated by that matrix. Both properties together say: on each momentum eigenspace the S^k form an $su(2)$ algebra commuting with momentum — precisely the structure needed to carry the $(2s+1)$ -dimensional spin space along every point of the mass shell. The Casimir consistency check: $\vec{S}^2 = -W^2/M^2 = s(s+1)$, using $W_{\text{rest}}^0 = 0$.

Helicity basis vs. spin-polarization basis at a glance.

	helicity basis $ M, \vec{p}, s, \lambda\rangle$	spin-pol. basis $ M, \vec{p}, s, \sigma\rangle$
standard transf.	$H(p) = \Lambda_R(\hat{z} \rightarrow \hat{p}) \Lambda_B(\vec{p} /M, \hat{z})$	$L(p) = \Lambda_B(\vec{p} /M, \hat{p})$ (direct boost)
label measures	$\vec{J} \cdot \hat{p}$ (along motion)	S^3 ($\hat{z}$ -axis of rest frame)
under rotations	λ invariant	$W_B(\Lambda_R, p) = \Lambda_R$: naive rotation of σ
natural use	massless limit, decays	spin addition, NR limit

6 Lorentz action in the spin–polarization basis [p. 40*–41*]

3) The states $|M, \vec{p}, s, \sigma\rangle$ transform under Lorentz as follows. [p. 41*]

$$U(\Lambda)|M, \vec{p}, s, \sigma\rangle = U(\Lambda) U_B(|\vec{p}|/M, \hat{p})|M, \vec{0}, s, \sigma\rangle = \dots$$

We introduce the transformation $U_B(|\vec{p}_\Lambda|/M, \hat{p}_\Lambda)$ such that, by definition,

$$U_B(|\vec{p}_\Lambda|/M, \hat{p}_\Lambda)|M, \vec{0}, s, \sigma\rangle = |M, \vec{p}_\Lambda, s, \sigma\rangle.$$

$$\dots = U_B(|\vec{p}_\Lambda|/M, \hat{p}_\Lambda) \left[U_B(|\vec{p}_\Lambda|/M, \hat{p}_\Lambda)^{-1} U(\Lambda) U_B(|\vec{p}|/M, \hat{p}) \right] |M, \vec{0}, s, \sigma\rangle \stackrel{\circledast}{=}$$

► this is an element of the little group, defined by

$$\Lambda_B(|\vec{p}_\Lambda|/M, \hat{p}_\Lambda)^{-1} \Lambda \Lambda_B(|\vec{p}|/M, \hat{p}) = W_B(\Lambda, p) \quad \text{Wigner rotation}$$

► in Weinberg notation:

$$W_B(\Lambda, p) = L(\Lambda p)^{-1} \Lambda L(p).$$

Remark: this is still a rotation, but it differs from $W(\Lambda, p)$ defined before in the helicity basis.

⊗ The rest of the computation is analogue to the one we did before, and we get

$$U(\Lambda)|M, \vec{p}, s, \sigma\rangle = \sum_{\sigma'} [D_s(W_B(\Lambda, p))]_{\sigma'\sigma} |M, \vec{p}_\Lambda, s, \sigma'\rangle$$

Added explanation (not in the original notes)

Same movie, different convention. The argument is word for word the Step-6 argument of Lecture 4 (p. 30*–32*): (i) $U(\Lambda)|M, \vec{p}, s, \sigma\rangle$ is an eigenstate of P^μ with eigenvalue $p_\Lambda = \Lambda p$; (ii) sandwiching $U(\Lambda)$ between the standard transformation reaching p_Λ and the one reaching p isolates a transformation that maps $p_{\text{ref}} \rightarrow p \rightarrow p_\Lambda \rightarrow p_{\text{ref}}$, hence an element of the little group $SO(3)$; (iii) that element acts on the rest states through the known spin- s matrices D_s . Only the *convention* for the standard transformation changed ($H \rightarrow L$), so the little-group element changes too: $W_B(\Lambda, p) \neq W(\Lambda, p)$ in general — they differ by the p -dependent rotations R_0 relating the two bases. The payoff of this particular convention is the beautiful property proved next.

7 Key property: $W_B(\Lambda_R, p) = \Lambda_R$ for pure rotations [p. 42*–45*]

[p. 42*] Comment. In the spin-polarization basis, the Wigner rotation enjoys a very important property:

if Λ is a pure rotation, $\Lambda = \Lambda_R$, we have $W_B(\Lambda_R, p) = \Lambda_R$

$\hookrightarrow$ Flavour of the proof.

The proof is based on an important structural property of the Lorentz group: the boost generators $\vec{K}$ form a vector, and transform as a vector under rotations:

$$\begin{aligned} D(R(\vec{\theta}))^{-1} K^i D(R(\vec{\theta})) &= R(\vec{\theta})_{ij} K^j \\ \Downarrow \text{ if we take } R(\vec{\theta})^{-1} \text{ instead of } R(\vec{\theta}) \\ D(R(\vec{\theta})) K^i D(R(\vec{\theta}))^{-1} &= (R(\vec{\theta})^{-1})_{ij} K^j = (R(\vec{\theta})^T)_{ij} K^j = (R(\vec{\theta}))_{ji} K^j \\ \Downarrow \text{ contract with some direction } \hat{p}^i \end{aligned}$$

[p. 43*] We thus get

$$D(R(\vec{\theta})) \hat{p} \cdot \vec{K} D(R(\vec{\theta}))^{-1} = (R(\vec{\theta})\hat{p}) \cdot \vec{K} \quad (*)$$

► transformation in a generic representation; $R(\vec{\theta})$ is the 3×3 matrix acting on vectors.

Consider now a rotation in the Lorentz group $\Lambda_R(\vec{\theta})$ and a boost $\Lambda_B(\eta, \hat{p})$. We construct

$$\begin{aligned} \Lambda_R(\vec{\theta}) \Lambda_B(\eta, \hat{p}) \Lambda_R(\vec{\theta})^{-1} &= \Lambda_R(\vec{\theta}) \exp(-i\eta \hat{p} \cdot \vec{K}) \Lambda_R(\vec{\theta})^{-1} \\ &= \exp\left[-i\eta \Lambda_R(\vec{\theta}) \hat{p} \cdot \vec{K} \Lambda_R(\vec{\theta})^{-1}\right]_{\text{using } C e^X C^{-1} = e^{CXC^{-1}}; \text{ we now use } (*)} \\ &= \exp\left(-i\eta (R(\vec{\theta})\hat{p}) \cdot \vec{K}\right) = \Lambda_B(\eta, R(\vec{\theta})\hat{p}). \end{aligned}$$

$$\Lambda_R(\vec{\theta}) \Lambda_B(\eta, \hat{p}) \Lambda_R(\vec{\theta})^{-1} = \Lambda_B(\eta, R(\vec{\theta})\hat{p})$$

► we now consider as rotation $\Lambda_R(\hat{z} \rightarrow \hat{p})$ and as boost $\Lambda_B(\eta, \hat{z})$; consequently, this is $R(\hat{z} \rightarrow \hat{p}) \hat{z} = \hat{p}$. We get

$$\Lambda_R(\hat{z} \rightarrow \hat{p}) \Lambda_B(\eta, \hat{z}) \Lambda_R(\hat{z} \rightarrow \hat{p})^{-1} = \Lambda_B(\eta, \hat{p})$$

Added explanation (not in the original notes)

Two remarks on this identity. (i) It says that the two conventions of Step 5 are explicitly related: $L(p) = \Lambda_B(|\vec{p}|/M, \hat{p}) = \Lambda_R(\hat{z} \rightarrow \hat{p}) \Lambda_B(|\vec{p}|/M, \hat{z}) \Lambda_R(\hat{z} \rightarrow \hat{p})^{-1}$, while $H(p) = \Lambda_R(\hat{z} \rightarrow \hat{p}) \Lambda_B(|\vec{p}|/M, \hat{z})$ — they differ exactly by the trailing rotation $\Lambda_R(\hat{z} \rightarrow \hat{p})^{-1}$, which fixes p_{ref} and is thus the little-group element $R_0(\vec{p})$ announced earlier: $L(p) = H(p) \Lambda_R(\hat{z} \rightarrow \hat{p})^{-1}$. (ii) The conjugation formula “rotate a boost and you get the boost along the rotated axis” is the finite-transformation version of the bracket $[J^i, K^j] = i\varepsilon_{ijk} K^k$ of Lecture 2 (p. 38): the commutator said *infinitesimally* that $\vec{K}$ is a vector under rotations; here the statement is exponentiated using $C e^X C^{-1} = e^{CXC^{-1}}$ (valid because conjugation commutes with every term of the power series — same trick as Lecture 2, p. 35).

[p. 44*] We now rewrite the very same equation for p_Λ instead of p . We also trade the rapidity for $|\vec{p}_\Lambda|/M$:

$$\Lambda_B(|\vec{p}_\Lambda|/M, \hat{p}_\Lambda) = \Lambda_R(\hat{z} \rightarrow \hat{p}_\Lambda) \Lambda_B(|\vec{p}_\Lambda|/M, \hat{z}) \Lambda_R(\hat{z} \rightarrow \hat{p}_\Lambda)^{-1}$$

► we now insert this equation in the definition of the Wigner rotation,

$$\Lambda_B(|\vec{p}_\Lambda|/M, \hat{p}_\Lambda)^{-1} \Lambda \Lambda_B(|\vec{p}|/M, \hat{p}) = W_B(\Lambda, p).$$

We get

$$\begin{aligned} W_B(\Lambda, p) &= \left[\Lambda_R(\hat{z} \rightarrow \hat{p}_\Lambda) \Lambda_B(|\vec{p}_\Lambda|/M, \hat{z}) \Lambda_R(\hat{z} \rightarrow \hat{p}_\Lambda)^{-1} \right]^{-1} \Lambda \underbrace{\Lambda_B(|\vec{p}|/M, \hat{p})}_{\text{also this term can be written using the previous equation}} \\ &= \Lambda_R(\hat{z} \rightarrow \hat{p}_\Lambda) \Lambda_B(|\vec{p}_\Lambda|/M, \hat{z})^{-1} \Lambda_R(\hat{z} \rightarrow \hat{p}_\Lambda)^{-1} \Lambda \Lambda_R(\hat{z} \rightarrow \hat{p}) \Lambda_B(|\vec{p}|/M, \hat{z}) \Lambda_R(\hat{z} \rightarrow \hat{p})^{-1}. \end{aligned}$$

We now specify the case in which $\Lambda = \Lambda_R$ is a pure rotation. We write [p. 45*]

$$W(\Lambda_R, p) = \Lambda_R(\hat{z} \rightarrow \hat{p}_\Lambda) \Lambda_B(|\vec{p}|/M, \hat{z})^{-1} \overbrace{\left(\Lambda_R(\hat{z} \rightarrow \hat{p}_\Lambda)^{-1} \Lambda_R \Lambda_R(\hat{z} \rightarrow \hat{p}) \right)}^{(*)} \Lambda_B(|\vec{p}|/M, \hat{z}) \Lambda_R(\hat{z} \rightarrow \hat{p})^{-1}$$

► this combination is a rotation around $\hat{z}$, since it leaves any vector along $\hat{z}$ invariant. As such, it commutes with boosts along $\hat{z}$:

$$\begin{aligned} &= \underbrace{\Lambda_R(\hat{z} \rightarrow \hat{p}_\Lambda) \Lambda_R(\hat{z} \rightarrow \hat{p}_\Lambda)^{-1}}_{\mathbb{I}} \Lambda_R \Lambda_R(\hat{z} \rightarrow \hat{p}) \underbrace{\Lambda_B(|\vec{p}|/M, \hat{z})^{-1} \Lambda_B(|\vec{p}|/M, \hat{z})}_{\mathbb{I}} \Lambda_R(\hat{z} \rightarrow \hat{p})^{-1} \\ &= \Lambda_R \underbrace{\Lambda_R(\hat{z} \rightarrow \hat{p}) \Lambda_R(\hat{z} \rightarrow \hat{p})^{-1}}_{\mathbb{I}} = \Lambda_R. \end{aligned}$$

(*) Since Λ is a rotation, $|\vec{p}_\Lambda| = |\vec{p}|$, and $\Lambda_B(|\vec{p}_\Lambda|/M, \hat{z})^{-1} = \Lambda_B(|\vec{p}|/M, \hat{z})^{-1}$.

Added explanation (not in the original notes)

Filling in the two key steps of p. 45*. (a) *Why is the combination (*) a rotation about $\hat{z}$?* Follow the unit vector $\hat{z}$ through the three factors: $\Lambda_R(\hat{z} \rightarrow \hat{p})$ maps $\hat{z} \rightarrow \hat{p}$; the rotation Λ_R maps $\hat{p} \rightarrow \hat{p}_\Lambda$ (for a pure rotation $\vec{p}_\Lambda = R\vec{p}$, so $\hat{p}_\Lambda = R\hat{p}$, and this is also why $|\vec{p}_\Lambda| = |\vec{p}|$, the fact used in the side remark (*) of the notes); finally $\Lambda_R(\hat{z} \rightarrow \hat{p}_\Lambda)^{-1}$ maps $\hat{p}_\Lambda \rightarrow \hat{z}$. Net effect: $\hat{z} \rightarrow \hat{z}$. A rotation whose axis contains $\hat{z}$ is precisely a rotation *about* $\hat{z}$, generated by J^3 alone. (b) *Why does it commute with boosts along $\hat{z}$?* At the level of generators, $[J^3, K^3] = i\varepsilon_{33k}K^k = 0$ (Lecture 2, p. 38); exponentiating, every rotation $e^{-i\alpha J^3}$ commutes with every boost $e^{-i\eta K^3}$. So the combination can be slid past $\Lambda_B(|\vec{p}|/M, \hat{z})^{-1}$, the two boosts annihilate each other, and then the rotations collapse pairwise to leave Λ_R .

Why this result is important. In the spin-polarization basis, rotations act on the spin index in the *naïve* way — $U(\Lambda_R)|M, \vec{p}, s, \sigma\rangle = \sum_{\sigma'} [D_s(\Lambda_R)]_{\sigma'\sigma} |M, R\vec{p}, s, \sigma'\rangle$, with the same $D_s(\Lambda_R)$ for every momentum. This means spins of different particles (different momenta) can be added with the ordinary Clebsch–Gordan rules, exactly as in non-relativistic QM: nothing momentum dependent contaminates the rotation properties. All the genuinely relativistic novelty is confined to *boosts*, for which $W_B(\Lambda, p)$ is a nontrivial p -dependent rotation — the Wigner/Thomas rotation, rooted in $[K^i, K^j] = -i\varepsilon_{ijk}J^k$ (see the Lecture 2 blue box on Thomas precession). Note also that in the transcription of p. 45* the notes momentarily drop the subscript B and write $W(\Lambda_R, p)$; it is of course the same W_B defined on p. 41*.

8 Unitary irreps of the Poincaré group: massless case [p. 1**]

Unitary irreps of the Poincaré group — massless case.

We consider

$$p^2 = 0 \quad \text{with} \quad p^0 = |\vec{p}| > 0.$$

Step 3: choice of the reference vector

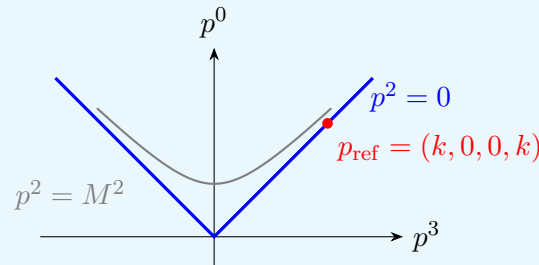
We consider

$$p_{\text{ref}}^\mu = (k, 0, 0, k), \quad k > 0; \quad P^\mu |0, k\hat{z}\rangle = p_{\text{ref}}^\mu |0, k\hat{z}\rangle, \quad P^2 |0, k\hat{z}\rangle = 0.$$

Added explanation (not in the original notes)

Where are Steps 1–2? They are unchanged from the massive case (Lecture 3, p. 1*–2*): Step 1 diagonalizes the commuting P^μ ; Step 2 classifies by the Casimir P^2 and, for $p^2 \geq 0$, by the invariant $\text{sign}(p^0)$. The massless case is the branch $p^2 = 0$, $p^0 > 0$ of that classification, so the construction resumes at Step 3. The state is written $|0, k\hat{z}\rangle$ following the notation $|M, \vec{p}\rangle$: mass 0, spatial momentum $\vec{p}_{\text{ref}} = k\hat{z}$. Note the crucial difference from the massive case: there is no rest frame — no Lorentz transformation can bring $(k, 0, 0, k)$ to the form $(M, \vec{0})$, because $p^2 = 0$ is invariant. The best one can do is to standardize the momentum to a fixed light-like vector along $\hat{z}$; the number $k > 0$ is a fixed reference energy (its precise value is irrelevant: any two choices are related by a boost along $\hat{z}$).

Added figure: the light cone in the (p^3, p^0) plane; the massless states live on the cone itself (blue), which for $p^0 > 0$ is the degenerate “mass shell” of the massive hyperbolas (grey, cf. Lecture 3, p. 2*).



Step 4: identification of the Wigner little group

We apply the theorem:

$$\vec{W} = \vec{J} + \frac{\vec{K} \times \vec{p}_{\text{ref}}}{p_{\text{ref}}^0} = \vec{J} + \frac{\vec{K} \times k\hat{z}}{k} = \vec{J} + \vec{K} \times \hat{z} = \vec{J} + K^2 \hat{x} - K^1 \hat{y}.$$

$\leadsto$ the three generators are

$$\begin{array}{ccc}
J^1 + K^2 \equiv T^1 \equiv T_x & & [T^1, T^2] = 0 \\
J^2 - K^1 \equiv T^2 \equiv T_y & \blacktriangleright & [J^3, T^1] = iT^2 \\
J^3 & & [J^3, T^2] = -iT^1
\end{array}
\quad \text{algebra of } SE(2)$$

[p. 2**]

$$G(p_{\text{ref}}^\mu) = SE(2) \quad \text{massless case.}$$

Added explanation (not in the original notes)

Worked check of the three brackets (using the Lorentz algebra of Lecture 2, p. 38: $[J^i, J^j] = i\varepsilon_{ijk}J^k$, $[J^i, K^j] = i\varepsilon_{ijk}K^k$, $[K^i, K^j] = -i\varepsilon_{ijk}J^k$):

$$\begin{aligned}
[T^1, T^2] &= [J^1 + K^2, J^2 - K^1] = \underbrace{[J^1, J^2]}_{iJ^3} - \underbrace{[J^1, K^1]}_0 + \underbrace{[K^2, J^2]}_0 - \underbrace{[K^2, K^1]}_{+iJ^3} = 0. \checkmark \\
[J^3, T^1] &= \underbrace{[J^3, J^1]}_{iJ^2} + \underbrace{[J^3, K^2]}_{-iK^1} = i(J^2 - K^1) = iT^2, \quad [J^3, T^2] = \underbrace{[J^3, J^2]}_{-iJ^1} - \underbrace{[J^3, K^1]}_{iK^2} = -iT^1. \checkmark
\end{aligned}$$

Compare with Lecture 1, p. 14: this is *exactly* the $se(2)$ algebra $[P^1, P^2] = 0$, $[J, P^1] = iP^2$, $[J, P^2] = -iP^1$, with the dictionary $T_x \leftrightarrow P^1$, $T_y \leftrightarrow P^2$, $J^3 \leftrightarrow J$: two commuting “translations” plus a “rotation” that rotates them into each other.

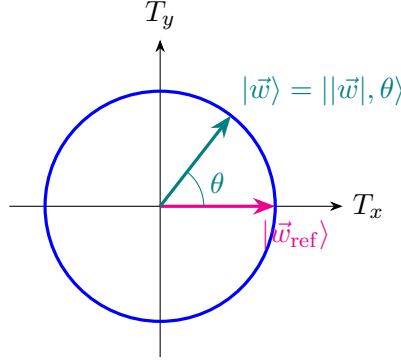
Geometric picture. The little group of a light-like vector is the Euclidean group of a plane, $ISO(2) \cong SE(2)$: the rotations about the momentum direction $\hat{z}$ (generated by J^3) obviously fix p_{ref} , but — unlike the massive case, where the stabilizer was the compact $SO(3)$ — they are joined by two *non-compact* combinations of rotations and boosts, T_x and T_y , called “null rotations”, which shear the light cone while keeping p_{ref} fixed. Consistency with the counting of the little-group theorem (Lecture 3, p. 7*): for any p , three generators. And here is the delightful payoff of the course’s structure: the group whose unitary irreps were patiently constructed in Lectures 1–2 as a “warm-up”, $SE(2)$, reappears as the little group governing the spin of *massless particles* — the warm-up was secretly Step 4 of the massless Poincaré problem.

9 Inducing from $SE(2)$: the continuous-spin dead end [p. 2**–3**]

As done in the massive case, we can use the irreps of the little group to “induce” irreps of the full Poincaré.

► we already constructed such irreps in the discussion about NRQM:

$$\begin{aligned}
&\left\{ |\vec{w}\rangle = U(\theta)|\vec{w}_{\text{ref}}\rangle, \theta \in [0, 2\pi) \right\}_{|\vec{w}|^2 \text{ fixed}} \quad (\star) \\
&T_x|\vec{w}\rangle = w_x|\vec{w}\rangle, \quad T_y|\vec{w}\rangle = w_y|\vec{w}\rangle, \quad (T_x^2 + T_y^2)|\vec{w}\rangle = (w_x^2 + w_y^2)|\vec{w}\rangle.
\end{aligned}$$



Following our logic, it seems appropriate to make the generalization: [p. 3**]

$$|0, k\hat{z}\rangle \xrightarrow{\text{“upgraded” to}} |0, k\hat{z}, |w|, \theta\rangle$$

► additional labels that describe the transformation under the little group.

△ Physical interpretation completely unclear. “0” is the mass, $k\hat{z}$ the momentum. We expect to find the spin, but we get a continuous quantum number, $|w|$. These are called indeed “continuous spin representations”.

We discard these representations.

It seems we arrived at a dead-end. However, there is an elegant way out. The point is that (★) does not exhaust all possible unitary irreps of $SE(2)$. We consider the special case in which we have

$$T_x|\vec{0}\rangle = 0, \quad T_y|\vec{0}\rangle = 0, \quad (T_x^2 + T_y^2)|\vec{0}\rangle = 0$$

$\rightsquigarrow$ in this particular case we can diagonalize simultaneously also J^3 .

Added explanation (not in the original notes)

Why the continuous-spin irreps are discarded. Group theory alone does not forbid them: for each $|w| > 0$ the construction of Lectures 1–2 gives a perfectly good unitary irrep of $SE(2)$, and inducing it up to Poincaré gives a consistent relativistic “particle”. The problem is phenomenological. Such a particle would carry, at fixed momentum, an *infinite* tower of internal states (the whole circle of $\vec{w}$ eigenvalues — an infinite-dimensional internal space, in stark contrast with the $(2s + 1)$ states of a massive particle), with a continuous invariant $|w|$ playing the role that the discrete spin plays elsewhere. No such particle has ever been observed: photons, gluons, gravitons all come with a *discrete* helicity and exactly one internal state per helicity value. This is why Wigner himself, after classifying them in 1939, set the continuous-spin representations aside, and the notes do the same. (They remain a topic of research — e.g. whether consistent interacting theories of “continuous spin particles” exist — but they play no role in the Standard Model.)

Why the zero-eigenvalue case allows diagonalizing J^3 too. In a generic irrep (★) one cannot diagonalize J^3 together with T_x, T_y : they do not commute, and J^3 moves the states around the circle. But suppose $T_x|\psi\rangle = T_y|\psi\rangle = 0$. Then the subspace annihilated by the “translations”

is *invariant* under J^3 : using $[T_x, J^3] = -iT_y$ (from the boxed algebra),

$$T_x (J^3 |\psi\rangle) = J^3 \underbrace{T_x |\psi\rangle}_0 + [T_x, J^3] |\psi\rangle = -i \underbrace{T_y |\psi\rangle}_0 = 0,$$

and similarly $T_y (J^3 |\psi\rangle) = 0$. So J^3 acts *within* that subspace and can be diagonalized there — exactly the special case the notes now exploit. Geometrically: the circle of radius $|w|$ degenerates to the single point $\vec{w} = \vec{0}$, which is left fixed by rotations; this is the $SE(2)$ analogue of the “ $p = 0$ ” irreps mentioned in the Lecture 2 blue box on p. 25, where translations act trivially and only the rotation label survives.

10 Degenerate irreps of $SE(2)$ and the helicity [p. 4**–5**]

[p. 4**] We thus write

$$T_x |\vec{0}, \lambda\rangle = 0, \quad T_y |\vec{0}, \lambda\rangle = 0, \quad J^3 |\vec{0}, \lambda\rangle = \lambda |\vec{0}, \lambda\rangle$$

- formally, this is the trivial representation under the translational part of $SE(2)$.
- What is λ ? Since we look for unitary representations, J^3 must be Hermitian and $\lambda \in \mathbb{R}$. There is actually more that we can say about λ . Consider a generic rotation,

$$D(\theta) = \exp(-i\theta J^3).$$

We thus have

$$D(\theta) |\vec{0}, \lambda\rangle = \exp(-i\theta J^3) |\vec{0}, \lambda\rangle = e^{-i\theta\lambda} |\vec{0}, \lambda\rangle.$$

In order for this transformation to be compatible with the group operation, we must have $D(\theta = 2\pi) = \mathbb{1}$, since a rotation of 2π is the identity:

$$D(\theta = 2\pi) = e^{-2\pi i\lambda} \stackrel{!}{=} 1 \quad \blacktriangleright \quad \lambda \in \mathbb{Z}.$$

- If we act on a state $|\vec{0}, \lambda\rangle$ with a generic $SE(2)$ element, we get [p. 5**]

$$\underbrace{e^{-i\alpha T_x - i\beta T_y}}_{\text{2D translation}} \underbrace{e^{-i\theta J^3}}_{\text{rotation}} |\vec{0}, \lambda\rangle = e^{-i\alpha T_x - i\beta T_y} e^{-i\theta\lambda} |\vec{0}, \lambda\rangle = e^{-i\theta\lambda} \underbrace{e^{-i\alpha T_x - i\beta T_y} |\vec{0}, \lambda\rangle}_{\text{reduces to the identity}} = e^{-i\theta\lambda} |\vec{0}, \lambda\rangle.$$

The action of a generic group element does not change the λ -eigenvalue.

- The state $|\vec{0}, \lambda\rangle$ with λ fixed forms a one-dimensional unitary irrep of $SE(2)$ in which the translation part of the group acts trivially $\rightsquigarrow$ a.k.a. “degenerate” unitary irrep of $SE(2)$. Rotations, on the contrary, act as

$$U(\theta) |\vec{0}, \lambda\rangle = e^{-i\theta\lambda} |\vec{0}, \lambda\rangle, \quad U(\theta + \phi) = U(\theta) U(\phi).$$

- Following our discussion on projective representations, we do include also half-integer values of λ :

$$\lambda \in \tfrac{1}{2}\mathbb{Z}.$$

Added explanation (not in the original notes)

The 2π argument, revisited. The step “ $D(2\pi) \stackrel{!}{=} \mathbb{1}$ hence $\lambda \in \mathbb{Z}$ ” is the one-dimensional version of the discussion of Lecture 4 (p. 14*–15*): there, $D_j(2\pi, \hat{z}) = (-1)^{2j} \mathbb{1}$ showed that half-integer j gives a *projective* representation (a representation up to a phase), and the QM argument — physical states are rays, defined up to a global phase — showed that projective representations are perfectly acceptable, being ordinary representations of the double cover ($SU(2)$ for $SO(3)$). Exactly the same logic applies to the rotation subgroup $SO(2) \subset SE(2)$: insisting on a true representation forces $e^{-2\pi i \lambda} = 1$, i.e. $\lambda \in \mathbb{Z}$; allowing the physically harmless phase $e^{-2\pi i \lambda} = \pm 1$ (i.e. passing to the double cover) admits $\lambda \in \frac{1}{2}\mathbb{Z}$, which is the choice made in the last yellow box. That is how a photon ($\lambda = \pm 1$) and a hypothetical graviton ($\lambda = \pm 2$) coexist in the same framework with spin- $\frac{1}{2}$ massless fermions ($\lambda = \pm \frac{1}{2}$, e.g. neutrinos in the massless approximation). One conceptual difference from the massive case is worth flagging: for $SO(3)$ the half-integer possibilities were rigidly organized by the algebra ($2j+1$ states, ladder operators, $j(j+1)$); for $SO(2)$, which is abelian, each irrep is one-dimensional and λ is *a priori* any real number — it is only the global (topological) requirement about 2π rotations that quantizes it. Continuity of representations of the double cover then allows precisely half-integers, not other fractions: going around the rotation circle twice must give back the identity, $e^{-4\pi i \lambda} = 1$.

11 The massless eigensystem [p. 6**]

We thus have the eigensystem: [p. 6**]

$$\begin{aligned} P^\mu |0, k\hat{z}, \lambda\rangle &= p_{\text{ref}}^\mu |0, k\hat{z}, \lambda\rangle \\ P^2 |0, k\hat{z}, \lambda\rangle &= 0 \\ J^3 |0, k\hat{z}, \lambda\rangle &= \lambda |0, k\hat{z}, \lambda\rangle, \quad \lambda = 0, \pm \frac{1}{2}, \pm 1, \dots \end{aligned}$$

► we suppress the extra $\vec{0}$ label, but it is understood that

$$T_x |0, k\hat{z}, \lambda\rangle = T_y |0, k\hat{z}, \lambda\rangle = 0.$$

► We introduce again $S_{\hat{z}}^\mu = (0, 0, 0, 1)$ with

$$-S_{\hat{z}}^\mu W_\mu = -W_3 = +W^3.$$

Since $\vec{W} = \vec{J}P^0 + \vec{K} \times \vec{P}$

$$\Rightarrow \vec{W} |0, k\hat{z}, \lambda\rangle = (\vec{J}k + k\vec{K} \times \hat{z}) |0, k\hat{z}, \lambda\rangle,$$

which implies

$$W^3 |0, k\hat{z}, \lambda\rangle = k J^3 |0, k\hat{z}, \lambda\rangle,$$

so that we have

$$\begin{aligned} P^\mu |0, k\hat{z}, \lambda\rangle &= p_{\text{ref}}^\mu |0, k\hat{z}, \lambda\rangle \\ P^2 |0, k\hat{z}, \lambda\rangle &= 0 \\ -\frac{S_{\hat{z}}^\mu W_\mu}{k} |0, k\hat{z}, \lambda\rangle &= \lambda |0, k\hat{z}, \lambda\rangle \end{aligned} \quad \boxed{\lambda \in \frac{1}{2}\mathbb{Z}}$$

Added explanation (not in the original notes)

Small check used silently: the third component of $\vec{K} \times \hat{z}$ vanishes, $(\vec{K} \times \hat{z})^3 = K^1(\hat{z})^2 - K^2(\hat{z})^1 = 0$ — only $T_x = (W^1\text{-part})$ and $T_y = (W^2\text{-part})$ contain the boosts, while $W^3 = k J^3$ on the reference states. This is just the component form of the little-group generators found on p. 1**: $\vec{W} = (k T_x, k T_y, k J^3)$ on the reference subspace, and the degenerate irrep sets the first two to zero.

What has been achieved, and the road ahead. The final yellow box is the massless counterpart of the massive rest-frame eigensystem of Lecture 4 (p. 20*), with a striking simplification: a massless particle carries *no spin multiplet at all* — just a single state per momentum, labelled by one number $\lambda \in \frac{1}{2}\mathbb{Z}$, the *helicity* (indeed $-S_{\hat{z}} \cdot W/k = J^3$ here measures the angular momentum along the direction of motion $\hat{z}$, exactly the quantity called helicity in the massive analysis). Three structural remarks, anticipating what presumably comes next: (i) On the degenerate states one finds $W^\mu |\dots\rangle = \lambda P^\mu |\dots\rangle$ (both are $(k\lambda)(1, 0, 0, 1)^T$ on the reference state, and the relation is covariant), so $W^2 = 0 = P^2$ and $W \cdot P = 0$: the second Casimir gives no new label, and the proportionality constant λ is the invariant that classifies the irrep. (ii) Because λ labels the one-dimensional little-group irrep itself — rather than a state *inside* a $(2s + 1)$ -multiplet as σ did — a Lorentz transformation cannot change it: for massless particles the helicity is Lorentz invariant, resolving the contrast with the massive case noted on p. 36*. States with $\pm\lambda$ (e.g. left/right circular polarization of light, $\lambda = \mp 1$) belong to *different* irreps of $ISO^\uparrow(1, 3)$, and are related only by parity — they are grouped together when (as for the photon) the interactions respect parity. (iii) The remaining steps mirror Steps 5–6 of the massive case: define $|0, \vec{p}, \lambda\rangle$ by a standard transformation taking $k\hat{z} \rightarrow \vec{p}$ (a boost along $\hat{z}$ that rescales $k \rightarrow |\vec{p}|$, then the rotation $\hat{z} \rightarrow \hat{p}$ of Lecture 2, p. 41), and compute the analogue of the Wigner rotation. Since the induced little-group element acts on a one-dimensional space, the massless transformation law will involve a mere *phase*, $U(\Lambda)|0, \vec{p}, \lambda\rangle = e^{-i\lambda\theta(\Lambda, p)}|0, \vec{p}_\Lambda, \lambda\rangle$, with $\theta(\Lambda, p)$ the rotation angle of the $SE(2)$ -valued Wigner element — presumably the business of the next lecture.

(End of Lecture 5 — the massless eigensystem at the reference momentum is complete; the notes are expected to continue with the states of generic momentum and the massless transformation law.)